mostly undisturbed, providing a sharp snapshot of the early universe. There are roughly 400 million of these primordial cosmic microwave photons streaming through every cubic meter of space, pristine relics of the early universe.

Initial measurements of the microwave background radiation revealed its temperature to be remarkably uniform, but as we discussed in Chapter 11, closer inspection, first achieved in 1992 by the Cosmic Background Explorer (COBE) and since improved by a number of observational undertakings, found evidence of small temperature variations, as illustrated in Figure 14.4a. The data are gray-scale coded, with light and dark patches indicating temperature variations of about a few ten-thousandths of a degree. The figure's splotchiness shows the minute but undeniably real unevenness of the radiation's temperature across the sky.

While an impressive discovery in its own right, the COBE experiment also marked a fundamental change in the character of cosmological research. Before COBE, cosmological data were coarse. In turn, a cosmological theory was deemed viable if it could match the broad-brush features of astronomical observations. Theorists could propose scheme after scheme with only minimal consideration for satisfying observational constraints. There simply weren't many observational constraints, and the ones that existed weren't particularly precise. But COBE initiated a new era in which the standards have tightened considerably. There is now a growing body of precision data with which any theory must reckon successfully even to be considered. In 2001, the Wilkinson Microwave Anisotropy Probe (WMAP) satellite, a joint venture of NASA and Princeton University, was launched to measure the microwave background radiation with about forty times COBE's resolution and sensitivity. By comparing WMAP's initial results, Figure 14.4b, with COBE's, Figure 14.4a, you can immediately see how much finer and more detailed a picture WMAP is able to provide. Another satellite, *Planck*, which is being developed by the European Space Agency, is scheduled for launch in 2007, and if all goes according to plan, will better WMAP's resolution by a factor of ten.